Figure 1: GRUB menu listing

```
net.ipv4.conf.default.send_redirects = 1
net.ipv4.conf.all.send_redirects = 0
net.ipv4.icmp_echo_ignore_broadcasts=1
net.ipv4.conf.default.forwarding=1
```

To load these parameters, run *sysctl -p*.

This completes the basic installation of OpenVZ and on the next reboot, you will have an OpenVZ server, which is ready to create virtual machines.

Creating your first container (virtual machine)

In order to create your first container, you need a template file. 'Templates' in OpenVZ refers to a compressed format of different operating systems that are available freely. You can browse the templates at *https://openvz.org/Download/template/precreated*.

Download your favourite operating system's template to the */vz/template/cache* directory and once completed, follow the steps shown in the next section.

The power of the shell comes in handy when creating containers or when amending the current settings related to a container. In this section, we will create our first container (virtual machine) and add a few parameters, including the IP address, so that you can access your machine over a LAN (or provide a public IP so that it can be accessible over a WAN as a standalone virtual server), name, DNS name server, *root* user password, etc. So, let's get started with creating the container first:

```
# vzctl create 101 --ostemplate centos-6-x86_64 --config basic
```

1. *vzctl*: This is the most useful command for OpenVZ for creating a new container, editing an IP address, changing the *root* password, etc. Any operation that you can think of relating to the container will use *vzctl*.
2. *--ostemplate*: This defines the template (operating system) you want to install in this new container space. All templates go in */vz/template/cache*.
3. *--config*: This will use the *ve-basic.conf-sample* configuration file present in the */etc/sysconfig/vz-scripts/* directory. Another option can be *-–config light* which will use the *ve-light.conf-sample* configuration file. You can check the contents of these files by opening them in your favourite text editor. There are other configuration files as well, by default, but I would recommend that you use one of the above just to keep things simple.

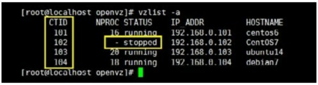

Figure 2: openvz_list_containers

After running the above command, you should see the following output along with some other information:

```
Creating container private area (centos-6-x86_64)
Unmounting file system at /vz/root/101
Unmounting device /dev/ploop15699
Opening delta /vz/private/101/root.hdd/root.hdd
Adding delta dev=/dev/ploop15699 img=/vz/private/101/root.
hdd/root.hdd (rw)
Mounting /dev/ploop15699p1 at /vz/root/101 fstype=ext4
data='balloon_ino=12,'
Performing postcreate actions
Unmounting file system at /vz/root/101
Unmounting device /dev/ploop15699
CT configuration saved to /etc/vz/conf/101.conf
Container private area was created
```

To check if your container was created successfully, run *vzlist -a* and it should list all the containers (even those not running) as shown in Figure 2.

Configuring the container (changing the IP address, DNS entry, name, etc)

Now that we have created our container, we need to provide an IP address, name, hostname, DNS name server, etc.

Adding the IP address to a container: This can be confusing because you need to provide an IP address which is accessible over the LAN, according to your set-up. If you have a pool of public IP addresses, then you can provide an IP from that range. For the sake of testing, check the IP address of your host machine (using *ifconfig*) and provide an IP from the same range. For example, if your host machine's IP address is 192.168.0.254, then use 192.168.0.253. Basically, you can use any unassigned IP address within the